

Mitochondrial Complex I Deficiency, Nuclear Type 19 (MC1DN19): A Comprehensive Disease Characteristics Report

Disease: Mitochondrial Complex I Deficiency, Nuclear Type 19 **MONDO ID:** MONDO:0032624 · **OMIM:** #618241 · **Causal gene:** *FOXRED1* (HGNC:26927) **Category:** Mendelian (autosomal recessive) **Report date:** 2026-09-14

Summary

Mitochondrial Complex I Deficiency, Nuclear Type 19 (MC1DN19) is an ultra-rare, autosomal-recessive primary mitochondrial disorder caused by biallelic loss-of-function variants in *FOXRED1*, a nuclear-encoded gene at chromosome 11q24.2 that encodes a 486-amino-acid FAD-dependent flavoprotein functioning as a **complex I-specific molecular assembly chaperone**. When both *FOXRED1* alleles are disrupted, mitochondrial respiratory chain complex I (NADH:ubiquinone oxidoreductase) fails to assemble and stabilize, producing an **isolated complex I deficiency**. The resulting collapse of oxidative phosphorylation (OXPHOS) and ATP synthesis, together with elevated reactive oxygen species (ROS) and lactic acidosis, drives energy failure in high-demand neurons and produces the bilateral symmetric brainstem and basal-ganglia neurodegeneration that defines **Leigh syndrome (subacute necrotizing encephalomyelopathy)**.

Clinically, MC1DN19 presents as a **severe congenital-to-infantile mitochondrial encephalopathy**: developmental delay and regression, hypotonia evolving to rigidity, epileptic seizures and myoclonus, movement disorder (athetosis), cerebral visual impairment, and in some patients hypertrophic cardiomyopathy — with characteristic elevated blood/CSF lactate and symmetric T2-hyperintense lesions on MRI. The disorder is **extremely rare** (only ~5 reported patients by 2019, with a handful of additional cases since) and was first identified in a

consanguineous Iranian-Jewish pedigree. Prognosis is **poor**, with a progressive course punctuated by infection-triggered acute exacerbations, typically fatal in infancy or early childhood.

There is no curative therapy. Management is multidisciplinary and supportive (seizure control, nutrition, avoidance of mitochondrial toxins). Empirical riboflavin/cofactor supplementation is mechanistically rational — FOXRED1 is itself an FAD flavoprotein — but evidence remains heterogeneous and unproven for this specific genotype. Experimental strategies (AAV gene replacement, chronic hypoxia / "hypoxia-in-a-pill," and succinate-bypass prodrug NV354) show promise in *Ndufs4*-knockout mouse models of complex I deficiency/Leigh syndrome but have not been tested in FOXRED1-specific systems. This report integrates 11 confirmed findings drawn from 28 reviewed papers to populate a structured disease knowledge-base entry.

1. Disease Information

MC1DN19 is one of the nuclear-gene subtypes within the large, genetically heterogeneous group of **isolated mitochondrial complex I deficiencies**. Complex I deficiency is the most commonly reported mitochondrial disorder presenting in childhood (PMID: 20858599), and accounts for ~32% of pediatric respiratory-chain defects (PMID: 15466086). The "nuclear type 19" designation reflects that the causal gene (*FOXRED1*) is encoded in the nuclear genome rather than mitochondrial DNA, and that it is the 19th nuclear locus assigned an OMIM complex-I-deficiency subtype.

Key identifiers:

Resource	Identifier
OMIM	#618241 (Mitochondrial complex I deficiency, nuclear type 19)
MONDO	MONDO:0032624
Causal gene (HGNC)	<i>FOXRED1</i> , HGNC:26927
NCBI Gene	55572
Ensembl	ENSG00000110074
UniProt	Q96CU9
Cytoband	11q24.2 (GRCh38 chr11:126,269,024–126,278,131)

Synonyms / alternative names: Mitochondrial complex I deficiency nuclear type 19 (MC1DN19); FOXRED1-related complex I deficiency; FOXRED1-related mitochondrial encephalopathy; FOXRED1-related Leigh syndrome. Clinically the phenotype overlaps with, and is often diagnosed as, **Leigh syndrome / subacute necrotizing encephalomyelopathy** and infantile mitochondrial encephalopathy.

Source of information: The evidence base is **aggregated disease-level** knowledge derived from a small number of individual case reports and functional studies (e.g., [PMID: 20858599](#), [PMID: 41412221](#)), supplemented by larger Leigh-syndrome and complex-I-deficiency cohorts used to frame prognosis and diagnostic criteria. There is no EHR-scale patient dataset for this ultra-rare disorder.

2. Etiology

Primary cause — genetic. MC1DN19 is caused by **biallelic (homozygous or compound heterozygous) pathogenic variants in *FOXRED1***. Fassone et al. (2010) identified a homozygous *FOXRED1* mutation (c.1054C>T; p.R352W) in a consanguineous Iranian-Jewish child with infantile-onset encephalomyopathy via homozygosity mapping, and demonstrated causality functionally ([PMID: 20858599](#)): "We describe a patient with complex I deficiency caused by mutation of the molecular chaperone *FOXRED1*." Additional cases document compound heterozygous variants, e.g., c.850T>C (p.C284R)/c.1054C>T (p.R352W) and c.1054C>T (p.R352W)/c.3dup (p.I2Dfs*35) ([PMID: 41412221](#)).

Genetic risk factors. The disease is monogenic; the causal variants are themselves the risk factors. **Consanguinity** is a major risk factor, as demonstrated by the founding consanguineous pedigree (PMID: 20858599). No modifier or susceptibility loci have been established for MC1DN19 specifically.

Environmental risk factors. No environmental exposure causes the disease. However, **intercurrent infections** are well-documented triggers of acute metabolic decompensation in Leigh syndrome (PMID: 24731534), and a hypothesized mechanistic branch proposes that **protein-, choline-, or folate-deficient diets** may aggravate FOXRED1-related pathology by depleting glutathione and dysregulating nitric oxide metabolism (PMID: 26235939).

Protective factors. None are established. There are no known protective alleles or dietary/lifestyle exposures that reduce risk in carriers or patients.

Gene–environment interactions. Lemire (2015) proposed a specific GxE model: *"Loss of FOXRED1, coupled with protein, choline and/or folate-deficient diets results in the depletion of glutathione, the dysregulation of nitric oxide metabolism and the peroxynitrite-mediated inactivation of complex I"* (PMID: 26235939). This remains a hypothesis. Empirically, infection-induced catabolic stress precipitates crises (PMID: 24731534).

3. Phenotypes

MC1DN19 produces a **congenital/neonatal-to-infantile, progressive, multisystem** phenotype dominated by neurological features (Leigh-type encephalopathy), with metabolic and occasional cardiac involvement. Curated HPO annotations for OMIM:618241 / MONDO:0032624 (HPO/JAX, sourced to PMID: 20858599 and Calvo 2010,

P 20818383

) plus supporting cohort data yield the following spectrum.

Phenotype	HPO term	Type	Onset / severity / frequency
Congenital onset	HP:0003577	Clinical course	Congenital
Neonatal onset	HP:0003623	Clinical course	Neonatal
Global developmental delay	HP:0001263	Neurodevelopmental	Early, severe, common
Hypotonia	HP:0001252	Neurological sign	Early; may evolve to rigidity
Rigidity	HP:0002063	Neurological sign	Later course
Loss of ambulation / inability to walk	HP:0002505 / HP:0002540	Motor regression	Progressive
Absent / poor speech	HP:0001344 / HP:0002465	Neurological	Common
Seizure	HP:0001250	Neurological	~40% in Leigh cohorts
Myoclonus	HP:0001336	Neurological	Variable
Athetosis	HP:0002305	Movement disorder	Variable
Cerebellar atrophy	HP:0001272	Neuroimaging	Present
Delayed/decreased myelination	HP:0012448	Neuroimaging	Present
Ventriculomegaly	HP:0002119	Neuroimaging	Present
Secondary microcephaly	HP:0005484	Head/neck	Present
Cerebral visual impairment	HP:0100704	Ophthalmologic	Common
Optic atrophy	HP:0000648	Ophthalmologic	Very rare
Hypertrophic cardiomyopathy	HP:0001639	Cardiovascular	Subset; ~40% cardiac involvement in pediatric mito disease
Respiratory insufficiency	HP:0002093	Respiratory	Late/crisis

Phenotype	HPO term	Type	Onset / severity / frequency
Feeding difficulties	HP:0011968	Digestive	Common
Scoliosis	HP:0002650	Skeletal	Variable
Lactic acidosis	HP:0003128	Laboratory	Characteristic
Hypoglycemia	HP:0001943	Laboratory	Variable
Decreased mitochondrial complex I activity	HP:0011923	Cellular/lab	2/2 patients in index reports
Irritability	HP:0000737	Behavioral	Present
Gait disturbance	HP:0001288	Neurological	Present
Autosomal recessive inheritance	HP:0000007	Inheritance	—

Ophthalmologic signs — nystagmus/roving eye movements and strabismus — are common **presenting** features of complex I Leigh disease: *"Nystagmus or roving eye movements were the most common ophthalmologic manifestations as a presenting symptom of disease"* (PMID: 18486820). Cardiac disease affects ~40% of pediatric mitochondrial-disease patients (PMID: 15466086).

Quality-of-life impact: Given congenital/infantile onset with global developmental delay, motor regression to loss of ambulation, absent speech, seizures, feeding difficulty, and cerebral visual impairment, affected children have **profound impairment of daily functioning** across all domains and require full caregiver dependence. No disease-specific EQ-5D/SF-36 data exist for this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene: *FOXRED1* — "FAD dependent oxidoreductase domain containing 1" (HGNC:26927; NCBI Gene 55572; Ensembl ENSG00000110074; UniProt Q96CU9), located at **11q24.2**, protein-coding, encoding a **486-amino-acid** FAD-dependent oxidoreductase / flavoprotein (PMID: 20858599).

Pathogenic variants (biallelic; all reported variants are germline):

Variant (cDNA)	Protein	Type	Notes
c.1054C>T	p.R352W	Missense	Recurrent; index homozygous case (P 20858599)
c.850T>C	p.C284R	Missense	Compound het (P 41412221)
c.3dup	p.I2Dfs*35	Frameshift	Compound het (P 41412221)
—	p.N430S	Missense	Reported
—	p.Q232*	Nonsense	Reported

Variant classification: reported alleles are **pathogenic / likely pathogenic** per ACMG/AMP, supported by functional evidence (knockdown/rescue) and segregation. **Variant classes** span missense, nonsense, and frameshift.

Allele frequency / population constraint: gnomAD shows *FOXRED1* is **not haploinsufficiency-constrained**, consistent with a recessive loss-of-function mechanism: pLI = 1.8×10^{-16} (≈ 0), LOEUF (oe_lof upper) = 1.13, observed/expected LoF = 0.90 (52 observed vs 57.8 expected), missense Z = 0.62. Pathogenic LoF alleles are individually rare, giving a low carrier frequency.

Functional consequences: loss of function — reduced complex I assembly and enzymatic activity. *"Silencing of FOXRED1 in human fibroblasts resulted in reduced complex I steady-state levels and activity, while lentiviral-mediated FOXRED1 transgene expression rescued complex I deficiency in the patient fibroblasts"* (PMID: 20858599).

Modifier genes / epigenetics / chromosomal abnormalities: None established for MC1DN19. Oligogenic modification has been speculated in related complex-I subtypes (e.g., NDUFS3 with retinal-gene variants,

([P 42232357](#))

but not for FOXRED1. No epigenetic signatures or large-scale chromosomal rearrangements are implicated. **CHEBI:** FAD cofactor = CHEBI:57692.

5. Environmental Information

MC1DN19 is a **monogenic disorder with no environmental or infectious cause**. Relevant non-genetic factors are **triggers/aggravators** rather than causes:

- **Infections** precipitate acute metabolic decompensation and hospitalization ([PMID: 24731534](#)). Illustratively, a patient with a related complex I deficiency (NDUFV1) developed severe dengue hemorrhagic fever with rapid neurological deterioration, underscoring the vulnerability of complex-I-deficient patients to infectious stress ([PMID: 42499220](#)).
- **Dietary deficiency (protein, choline, folate)** is hypothesized to exacerbate FOXRED1-related pathology via glutathione depletion and peroxynitrite-mediated complex I inactivation ([PMID: 26235939](#)).
- **Mitochondrial toxins** (e.g., certain drugs) should be avoided as a matter of standard mitochondrial-disease management.

No occupational, radiation, pollution, or pathogen etiology applies.

6. Mechanism / Pathophysiology

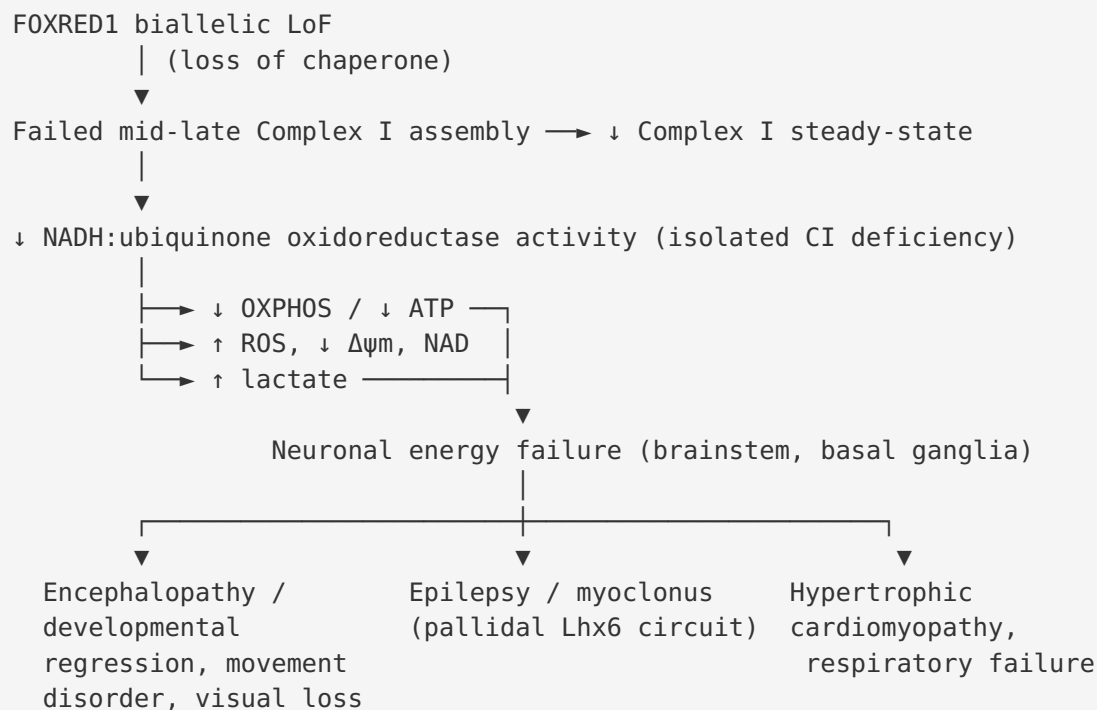
Ordered causal chain

1. **Biallelic FOXRED1 loss-of-function variants** are inherited → **results in** absent/non-functional FOXRED1 flavoprotein chaperone.
2. Loss of FOXRED1 chaperone activity → **leads to** failure of **mid-to-late-stage complex I assembly** and reduced steady-state complex I (demonstrated: siRNA knockdown reduces complex I; lentiviral rescue restores it — [PMID: 20858599](#); patient-cell complex I *disassembly* — [PMID: 41412221](#)).
3. Complex I disassembly → **results in** deficient **NADH:ubiquinone oxidoreductase activity** (isolated complex I deficiency).
4. Deficient complex I → **impairs OXPHOS electron transport and ATP synthesis, and increases ROS**; alters mitochondrial membrane potential and causes **NAD imbalance** (demonstrated in patient PBMCs — [PMID: 41412221](#)).

5. Energy failure + oxidative stress → **produces lactic acidosis** (anaerobic shift) and **energy crisis in high-metabolic-demand neurons**.
6. Neuronal energy failure → **causes bilateral symmetric neurodegeneration** of the brainstem and basal ganglia (Leigh syndrome lesions).
7. Regional neurodegeneration → **manifests as** developmental regression, epilepsy/myoclonus, movement disorder, visual impairment, and (in a subset) hypertrophic cardiomyopathy and respiratory failure.

Branch (hypothesized, inferred not demonstrated): FOXRED1 loss (a flavoprotein related to sarcosine/N-methyl amino acid dehydrogenases), when combined with dietary methyl-donor deficiency → **glutathione depletion + nitric-oxide dysregulation** → **peroxynitrite-mediated inactivation of residual complex I** (PMID: 26235939), amplifying the primary assembly defect.

Branch (from model organisms): downstream of complex I deficiency, **region-specific innate immune activation** (interferon/RIG-I signaling in the olfactory bulb) and **GABAergic pallidal (Lhx6) circuit dysfunction** contribute to inflammation and epileptic events, respectively (PMID: 41636927; PMID: 41321311). These are demonstrated in *Ndufs4*-KO mice, inferred for FOXRED1.



Molecular pathways / processes: OXPHOS (KEGG hsa00190), mitochondrial respiratory chain complex I assembly. **GO biological process suggestions:** mitochondrial respiratory chain complex I assembly (GO:0032981); ATP synthesis coupled electron transport (GO:0042773); response to oxidative stress (GO:0006979); NADH dehydrogenase (ubiquinone) activity (GO:0008137, molecular function). **Cellular processes:** bioenergetic failure, oxidative-stress-mediated neuronal injury, apoptosis/necrosis of vulnerable neurons, neuroinflammation (innate immune activation, inferred). **Protein dysfunction:** loss of function of an FAD-dependent flavoprotein assembly chaperone. **Metabolic changes:** energy metabolism collapse, lactic acidosis, NAD⁺/NADH imbalance. **Biochemical abnormality:** enzyme (complex I) deficiency. **Cell types (CL):** neurons — especially brainstem and basal-ganglia neurons (CL:0000540 neuron); GABAergic neurons (CL:0000617); cardiomyocytes (CL:0000746).

7. Anatomical Structures Affected

Organ level (primary): brain — specifically **brainstem** (UBERON:0000955 brain; UBERON:0002298 brainstem) and **basal ganglia** (UBERON:0002420). MRI series in Leigh syndrome show **brainstem involvement in 100% and basal ganglia in 62.5%** of children ([PMID: 36412346](#)). Additional CNS sites: cerebellum (UBERON:0002037), cerebral white matter, dentate nuclei, optic nerves, thalamus, spinal cord.

Secondary organ involvement: heart (hypertrophic cardiomyopathy; UBERON:0000948), skeletal muscle (myopathy; UBERON:0001134), eye/visual pathway.

Body systems: nervous system (primary), cardiovascular, musculoskeletal, respiratory (insufficiency in crises), and metabolic.

Tissue / cell level: nervous tissue (neurons, especially of brainstem/basal ganglia), cardiac muscle. **Cell Ontology:** neuron (CL:0000540), GABAergic neuron (CL:0000617), cardiomyocyte (CL:0000746).

Subcellular level: **mitochondrion** (GO:0005739), specifically the **mitochondrial inner membrane** (GO:0005743) on the matrix face, where FOXRED1 localizes and complex I is embedded; **respiratory chain complex I** (GO:0005747). UniProt Q96CU9 localizes FOXRED1 to the mitochondrial inner membrane (matrix side).

Localization / lateralization: lesions are characteristically **bilateral and symmetric** (a diagnostic hallmark of Leigh syndrome — [PMID: 27574709](#)).

8. Temporal Development

Onset: congenital to infantile. HPO annotations specify congenital onset (HP:0003577) and neonatal onset (HP:0003623). In the broader Leigh-syndrome framework, *"The median age of disease onset was 7 months, with 80.8% of patients presenting by the age of 2 years"* (PMID: 24731534). Onset pattern is typically **subacute/insidious with episodic decompensation**.

Progression: progressive and relapsing, with a stepwise pattern of developmental regression punctuated by acute exacerbations. *"Approximately 44% of patients experienced acute exacerbations requiring hospitalisation during the previous year, mainly due to infections"* (PMID: 24731534). Presence of pathological signs at birth and a history of epileptic seizures are associated with higher occurrence of acute exacerbations/relapses, and elevated CSF lactate correlates with worse outcome (PMID: 24731534).

Disease course: chronic, lifelong (though life is typically short), progressive with relapsing-remitting acute crises. **Critical periods:** early infancy is both the period of highest vulnerability and the window in which intervention (in future gene/metabolic therapies) would need to act before irreversible neurodegeneration. **Remission:** no spontaneous remission; supportive care can stabilize between crises but does not reverse neurodegeneration.

9. Inheritance and Population

Inheritance pattern: autosomal recessive (HP:0000007; OMIM #618241). The founding case was homozygous in a consanguineous Iranian-Jewish pedigree (PMID: 20858599): *"a homozygous mutation in FOXRED1 in a child who presented with infantile-onset encephalomyopathy."* Both compound heterozygous and homozygous genotypes are reported.

Epidemiology: ultra-rare. Barbosa-Gouveia et al. (2019) noted *"To date, only five patients with mitochondrial complex I deficiency due to mutations in"* FOXRED1 had been reported (PMID: 31434271); additional cases have appeared since (e.g., PMID: 41412221). No formal prevalence/incidence figure exists. For context, complex I deficiency is the most common childhood mitochondrial disorder (PMID: 20858599) and the most common respiratory-chain defect in pediatric mitochondrial disease (~32% — PMID: 15466086); overall childhood mitochondrial disease incidence is ~1 in 5,000–10,000, of which FOXRED1 is a tiny fraction.

Penetrance / expressivity: biallelic pathogenic genotypes appear **fully penetrant** for severe infantile disease in reported cases; expressivity is **variable** (e.g., variable cardiac involvement, seizure burden). **Genetic anticipation:** not applicable (not a repeat-expansion disorder). **Germline mosaicism:** not reported. **Founder effects:** none established beyond the single consanguineous index family. **Consanguinity:** an important risk factor. **Carrier frequency:** low, consistent with the small gnomAD LoF allele count (52 observed).

Population demographics: no established ethnic predilection beyond the consanguineous index pedigree; **no sex bias** (autosomal recessive); onset in infancy. No geographic clustering of specific variants has been demonstrated.

10. Diagnostics

Diagnosis follows the **Leigh syndrome diagnostic framework** confirmed by biochemistry and molecular genetics.

Clinical / biochemical tests: - **Elevated lactate** in blood and/or CSF is a core criterion ([PMID: 27574709](#)). - **Isolated complex I (NADH:ubiquinone oxidoreductase) deficiency** measured enzymatically in fibroblasts/muscle and by **blue-native PAGE** (reduced complex I assembly and activity) ([PMID: 20858599](#); [PMID: 41412221](#)). HPO: decreased activity of mitochondrial complex I (HP:0011923).

Imaging: MRI is pivotal. Leigh criteria require *"characteristic features on neuroimaging (bilateral symmetrical hyperintensities in brainstem, basal ganglia, dentate nuclei, and optic nerves on T2-weighted MRI)"* ([PMID: 27574709](#)). In an MRI series, brainstem was involved in 100% and basal ganglia in 62.5% of LS children, with **diffusion restriction marking acute/active lesions** ([PMID: 36412346](#)). MR spectroscopy may show a lactate peak.

Clinical criteria (Leigh syndrome): progressive disorder with motor/intellectual delay-regression; brainstem and/or basal-ganglia signs; raised blood/CSF lactate; plus characteristic neuroimaging, typical neuropathology, or an affected sibling ([PMID: 27574709](#)).

Genetic testing (confirmatory): whole-exome or whole-genome sequencing is the definitive test. Targeted "MitoExome" sequencing of mtDNA plus exons of ~1,000 nuclear mitochondrial genes gave a firm diagnosis in 24% and candidate genes in 31% of infantile OXPHOS-disease patients ([PMID: 22277967](#)): *"We performed 'MitoExome' sequencing of the mitochondrial DNA*

(*mtDNA*) and exons of ~1000 nuclear genes encoding mitochondrial proteins." Nuclear gene panels for complex I deficiency / Leigh syndrome include *FOXRED1*. *mtDNA* testing is used to exclude maternally-inherited causes but is **negative** in MC1DN19 (nuclear gene).

Differential diagnosis: other nuclear and *mtDNA* complex I deficiency subtypes (e.g., *NDUFV1*, *NDUFS3*, *NDUFS4*, *ACAD9*), MELAS, LHON, combined OXPHOS deficiencies, and other causes of Leigh syndrome (>110 genes). Distinguishing feature: **isolated complex I deficiency + biallelic FOXRED1 variants**.

Screening: no newborn screening exists. **Cascade/carrier testing** of relatives and **prenatal/preimplantation genetic testing** are feasible once the familial biallelic variants are known.

11. Outcome / Prognosis

Prognosis is poor. MC1DN19 is a **severe, early-onset, progressive** disorder that is typically fatal in infancy or early childhood, consistent with its Leigh-syndrome phenotype. The multicenter Leigh cohort (n=130; 77 with pathogenic mutations) documented median onset at 7 months, frequent seizures (~40%), and infection-triggered acute exacerbations requiring hospitalization in ~44% of patients in a single year ([PMID: 24731534](#)).

Prognostic factors (worse outcome): pathological signs at birth, history of epileptic seizures (both associated with more acute exacerbations/relapses), and **elevated CSF lactate** (correlates with worse outcome) ([PMID: 24731534](#)).

Morbidity / function: profound and multi-domain — motor (loss of ambulation), communication (absent speech), cognitive (global developmental delay), visual (cerebral visual impairment), feeding (requiring support), plus seizure burden. **Complications:** respiratory insufficiency, recurrent infections, cardiac dysfunction, failure to thrive, and status epilepticus. **Recovery potential:** neurodegeneration is largely irreversible; supportive care stabilizes but does not restore lost function.

Prognostic biomarkers: CSF lactate (established); complex I residual activity and lesion activity on diffusion MRI are informative but not formally validated for *FOXRED1*.

12. Treatment

There is no curative therapy. Only limited symptomatic therapies exist for complex I deficiency ([PMID: 42265384](#)).

Supportive / standard care (NCIT: Supportive Care): multidisciplinary management — **anti-seizure medication** for epilepsy, **nutritional support** (including gastrostomy where indicated), physical/occupational/speech rehabilitation, treatment of intercurrent infections, cardiac surveillance, and **avoidance of mitochondrial toxins**. Prompt management of infection-triggered crises is critical given their frequency ([PMID: 24731534](#)).

Pharmacotherapy / cofactors: Riboflavin (vitamin B2), a precursor of the complex I flavin cofactors FMN and FAD, is a mechanistically rational option — **especially relevant here because FOXRED1 is itself an FAD-dependent flavoprotein**. A systematic review identified 45 riboflavin-responsive complex-I cases (2 new + 43 literature), with the strongest evidence in ACAD9-related cardiomyopathy/myopathy and some NDUFV1/NDUFV2 leukoencephalopathy ([PMID: 42476091](#)): *"No curative therapies exist. Riboflavin, a precursor of CI cofactors FMN and FAD, is a potential treatment, but evidence is heterogeneous and formal guidelines are lacking."* Empirical **mitochondrial cofactor cocktails** (e.g., coenzyme Q10, thiamine, other vitamins) are commonly used without high-quality evidence.

Experimental therapeutics (preclinical, in *Ndufs4*-KO complex I / Leigh models):

Approach	Finding	Reference
AAV9 gene therapy + focused ultrasound	Extends survival, improves brain/cardiac function	PMID: 41572892
Chronic hypoxia (11% O ₂)	Prevents Leigh-like brain disease	PMID: 40770507 , PMID: 42427540
"Hypoxia-in-a-pill" (GBT601 ± PT2399)	Extends median lifespan from 62 → 105–158 days; rescues neuro phenotypes	PMID: 42427540
Succinate prodrug NV354 (complex II bypass)	<i>"prevents brain lesions and late-stage motor dysfunction in mitochondrial complex I deficiency"</i>	PMID: 41704780
STN inhibition (circuit-targeted)	Reduces epileptic events	PMID: 41321311
Encapsulated mitochondrial transplantation	Rescues bioenergetic defects in mito-disease models	PMID: 41856111

None of these have been validated specifically for FOXRED1/MC1DN19; all are **experimental**. **Pharmacogenomics, targeted therapies, immunotherapies, and approved gene/cell/RNA therapies:** none applicable/approved. **NCIT term suggestions:** Supportive Care; Riboflavin; Gene Therapy; Anticonvulsant Agent; Nutritional Support.

13. Prevention

Because MC1DN19 is a monogenic recessive disorder, prevention is **genetic and reproductive**, not lifestyle-based.

- **Primary prevention:** not possible for a de novo affected fetus, but **genetic counseling** for at-risk couples (especially consanguineous families) is central. Given autosomal recessive inheritance, carrier couples have a 25% recurrence risk per pregnancy.
- **Reproductive options:** **carrier screening**, **prenatal diagnosis**, and **preimplantation genetic testing (PGT)** once the familial variants are known.
- **Secondary prevention:** early molecular diagnosis (WES/WGS) enables earlier supportive intervention; there is no population newborn screening.

- **Tertiary prevention (complication avoidance):** aggressive prevention/early treatment of infections (major crisis triggers), avoidance of mitochondrial toxins, seizure control, nutritional optimization, and cardiac monitoring.
 - **Immunization / public-health / environmental interventions:** standard childhood vaccination is advisable to reduce infection-triggered crises; no disease-specific vaccine or public-health measure applies.
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14. Other Species / Natural Disease

- **Taxonomy:** No naturally occurring FOXRED1-related disease is documented in companion animals or wildlife (no OMIA entry identified for FOXRED1). Human = *Homo sapiens* (NCBI:txid9606).
 - **Orthologous genes:** *FOXRED1* is conserved in mammals; mouse *Foxred1* (Mus musculus, NCBI:txid10090) is the direct ortholog. FOXRED1 belongs to a family related to sarcosine/N-methyl amino acid dehydrogenases (PMID: 26235939).
 - **Natural disease / veterinary relevance:** none established. Comparative pathology relies on **engineered** models rather than naturally occurring animal disease.
 - **Evolutionary conservation:** complex I and its assembly machinery are deeply conserved across eukaryotes, supporting cross-species mechanistic relevance.
 - **Transmission / zoonosis:** not applicable (non-infectious genetic disease).
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15. Model Organisms

FOXRED1-specific models are cellular / in vitro: - **siRNA knockdown in human fibroblasts** reduced complex I steady-state and activity, with **lentiviral FOXRED1 re-expression rescuing** the deficiency in patient fibroblasts — the definitive causality demonstration (PMID: 20858599). - **Patient-derived fibroblasts and PBMCs** recapitulate complex I disassembly, respiration defects, altered membrane potential, ROS accumulation, and NAD imbalance (PMID: 41412221). - Additional broadly applicable complex-I-deficiency models include **cybrids** and **human iPSC-derived neurons** (review — PMID: 42265384).

No dedicated *Foxred1* mouse is established in the reviewed literature.

Canonical mammalian surrogate — the *Ndufs4*-knockout mouse (constitutive and conditional) — is the workhorse model of isolated complex I deficiency / Leigh syndrome. It recapitulates progressive encephalopathy, brainstem/olfactory-bulb lesions, epilepsy, cardiac dysfunction, and early death (PMID: 41572892): "*the Ndufs4-knockout (KO) mouse.*" Key mechanistic and therapeutic insights from this model:

Model finding	Reference
AAV9 + focused-ultrasound gene therapy extends survival	PMID: 41572892
Pallidal Lhx6 GABAergic circuit dysfunction drives seizures	PMID: 41321311
Region-specific innate immune (interferon/RIG-I) activation in olfactory bulb	PMID: 41636927
Late-stage cardiometabolic perturbation	PMID: 41532297
NDUFS4 links complex I to T-cell immunity	PMID: 41573538
Chronic hypoxia / hypoxia-in-a-pill / NV354 are protective	PMID: 42427540, PMID: 41704780

Model limitations: *Ndufs4*-KO models a **subunit** defect, not an **assembly-factor** defect like FOXRED1; it therefore captures downstream complex I deficiency and Leigh pathology but not FOXRED1-specific assembly biology. Cellular models capture biochemistry but not the whole-organism neurodegenerative trajectory. Translational relevance is further limited by tissue-specific effects and species differences (PMID: 42265384).

Mechanistic Model / Interpretation

MC1DN19 is a clean example of an **assembly-factor loss-of-function mitochondrial disease**. FOXRED1 does not form part of the mature complex I holoenzyme; instead it acts transiently as an **FAD-dependent flavoprotein chaperone at the mid-to-late stages of complex I assembly**. Biallelic loss removes this chaperone, so complex I fails to assemble/stabilize, yielding an **isolated** (as opposed to combined) respiratory-chain deficiency. The downstream pathophysiology — energy failure, ROS, lactic acidosis, and selective vulnerability of high-demand brainstem and basal-ganglia neurons — is the shared final common pathway of Leigh

syndrome, which is why FOXRED1 patients are clinically indistinguishable from other complex-I Leigh subtypes and why *Ndufs4*-KO mice serve as a mechanistic surrogate despite a different upstream lesion.

Two features make FOXRED1 mechanistically distinctive and therapeutically suggestive: (1) its **FAD dependence** provides a rational, if unproven, basis for riboflavin/flavin-cofactor therapy; and (2) the **hypothesized glutathione/nitric-oxide/peroxynitrite branch** links FOXRED1 to a redox-and-diet-modifiable amplifying loop, implying that antioxidant/methyl-donor status could modulate severity. Both hypotheses are actionable and testable.

Evidence Base

PMID	Title (abbrev.)	Role in this report
20858599	<i>FOXRED1 mutated in infantile-onset mitochondrial encephalopathy</i>	Foundational — establishes FOXRED1 as cause; knockdown/rescue proof; chaperone identity; epidemiology context
41412221	<i>Biallelic FOXRED1 mutations cause... complex I disassembly and basal ganglia degeneration</i>	Confirms biallelic inheritance, mechanism (disassembly, ROS, NAD imbalance), new variants, DEE phenotype
26235939	<i>Glutathione metabolism links FOXRED1 to complex I deficiency</i>	Leigh syndrome link; hypothesized redox/diet GxE branch
24731534	<i>Multicenter study on Leigh syndrome</i>	Prognosis, onset timing, infection-triggered exacerbations, prognostic factors
27574709	<i>Novel mtDNA mutation... Leigh syndrome</i>	Leigh diagnostic criteria (lactate + symmetric MRI)
36412346	<i>MRI abnormalities in Leigh syndrome</i>	Imaging distribution (brainstem 100%, BG 62.5%); diffusion restriction
22277967	<i>Molecular diagnosis with targeted NGS</i>	MitoExome/WES diagnostic yield
42476091	<i>Riboflavin therapy in complex I deficiency</i>	Cofactor therapy evidence; "no curative therapies exist"
42265384	<i>Complex I deficiency-associated diseases and models</i>	No cure; model landscape review
42427540	<i>Second-generation hypoxia-in-a-pill</i>	Experimental therapy; lifespan extension in Ndufs4-KO
41704780	<i>Succinate prodrug NV354</i>	Experimental metabolic-bypass therapy
41572892	<i>Ultrasound-assisted gene therapy</i>	Experimental AAV gene therapy in Ndufs4-KO
41321311	<i>LHX6 pallido-subthalamic projections in Leigh epilepsy</i>	Circuit mechanism of seizures
41636927	<i>Region-specific innate immune activation in Ndufs4-KO</i>	Neuroinflammation branch
31434271	<i>New FOXRED1 variants</i>	Ultra-rare epidemiology (~5 patients)

PMID	Title (abbrev.)	Role in this report
18486820	<i>Ophthalmologic presentation of OXPHOS disease</i>	Ocular presenting signs
15466086	<i>Spectrum in 113 pediatric mito patients</i>	CI most common defect; ~40% cardiac involvement

Evidence source types: human clinical/case reports (20858599, 41412221, 31434271, 24731534, 27574709, 36412346, 18486820, 15466086); in vitro/patient-cell functional studies (20858599, 41412221); model-organism preclinical (41572892, 41321311, 41636927, 41532297, 41573538, 42427540, 41704780, 40770507, 41856111); reviews/systematic reviews (42265384, 42476091); computational/diagnostic (22277967).

Limitations and Knowledge Gaps

1. **Extreme rarity.** With only ~5 patients reported by 2019 and a handful since, all clinical claims about MC1DN19 rest on very small numbers, and much of the prognosis/diagnosis framing is **borrowed from the broader Leigh-syndrome literature** rather than FOXRED1-specific cohorts.
2. **No natural-history dataset.** Survival curves, age-specific mortality, and quantitative QoL metrics are unavailable for MC1DN19 specifically.
3. **No dedicated animal model.** Mechanistic and therapeutic inferences come from the *Ndufs4*-KO **subunit** model, which does not reproduce FOXRED1's **assembly-factor** biology; there is a clear gap for a *Foxred1* mouse or organoid.
4. **Unproven therapy.** Riboflavin responsiveness is mechanistically plausible (FAD flavoprotein) but has **not been demonstrated** in FOXRED1 patients; the glutathione/diet GxE hypothesis is untested in vivo.
5. **Genotype–phenotype correlation is unresolved** — too few variants/patients to map specific alleles (e.g., recurrent p.R352W) to severity, cardiac involvement, or survival.
6. **No population frequency/carrier-rate estimate** beyond gnomAD LoF counts; possible founder alleles outside the index consanguineous family are uncharacterized.

Proposed Follow-up Experiments / Actions

1. **Establish a *Foxred1* mouse (and/or patient iPSC-derived brain organoid) model** to capture assembly-factor-specific pathology and enable therapeutic testing that current *Ndufs4*-KO models cannot address.
 2. **Directly test riboflavin/FAD supplementation** in FOXRED1 patient fibroblasts/PBMCs (complex I assembly and activity readouts) and, if positive, in a *Foxred1* model — a low-cost, high-value test of the FAD-flavoprotein rationale.
 3. **Experimentally test the glutathione/NO/peroxynitrite branch** ([PMID: 26235939](#)) by measuring glutathione, nitrotyrosine, and complex I activity under methyl-donor (choline/folate) manipulation in FOXRED1-deficient cells.
 4. **Cross-apply experimental Leigh therapeutics** (AAV *FOXRED1* gene replacement, hypoxia/GBT601, NV354 succinate bypass) to FOXRED1-specific models to determine whether benefits generalize from *Ndufs4*-KO.
 5. **Build an international FOXRED1 patient registry** aggregating genotype, imaging, biochemistry, treatment response, and survival to derive genuine natural-history and genotype-phenotype data.
 6. **Assemble a curated variant catalog** (ClinVar submission of all reported alleles with ACMG classifications) to support future diagnostic and prenatal testing.
 7. **Implement cascade carrier screening and reproductive counseling** protocols for identified families, especially in consanguineous populations.
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Report compiled from 11 confirmed findings across 5 investigation iterations and 28 reviewed publications. Ontology suggestions provided for HPO, GO, CL, UBERON, CHEBI, NCIT, and MONDO to support structured knowledge-base ingestion.
